SHAKTI PRASAD MOHAPATRA

Software Engineer-Quality Assurance | Payments and FinTech | EMV, ISO-8583, L2 Kernel and L3 Card Certification

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PROFESSIONAL SUMMARY

Senior QA Engineer with four years in payments, covering POS applications, settlement and reconciliation, terminal management and automated fuel dispensing. Most of that has been front-line work for tier-1 clients: release testing for Gilbarco across more than 500 Costco fuelling sites in North America, and end-to-end ownership of Geidea's settlement and reconciliation platform in Saudi Arabia. Certification is the specialism, with L2 kernel and L3 card applications approved for Visa, Mastercard, Amex, UnionPay and Diners Club, every scheme cleared inside two submission cycles. The rest of the day-to-day is REST API automation in Python and Selenium, log-level defect triage across host, acquirer and terminal layers, and, on one engagement, running a QA team of four. Now applying that same evidence-first discipline to LLM behaviour in regulated financial workflows.

TECHNICAL SKILLS

Payments and Standards: EMV covering contact, contactless and magnetic stripe, ISO-8583 message parsing down to MTIs, DE fields and bitmap debugging, ISO 20022, NEXO, L2 Kernel and L3 Card Certification for Visa, Mastercard, Amex, UnionPay and Diners Club, PCI DSS and 3DS, working knowledge of Base24

Platforms Tested: POS applications, settlement and reconciliation, terminal management systems, merchant onboarding with MID and TID provisioning, automated fuel dispensers, mobile payment apps on iOS and Android

Automation and Scripting: Python, Selenium WebDriver, pytest, REST API automation, Postman, shell scripting on Linux, Git and GitHub

Terminal Hardware: Verifone, Geidea, Lavego, Pine Labs, Gilbarco SmartCRIND on GSOM and Omnia boards

Tools: UL Certification Tool, ICC Tool, host simulators, JIRA, Zephyr, TestRail, Confluence, DBeaver and SQL, Jenkins, Docker, WinSCP, PuTTY

Test Disciplines: Functional, integration and end-to-end, regression, smoke and sanity, UAT, database validation, log analysis across multiple layers, root-cause analysis, Agile and Scrum

LLM Evaluation: promptfoo, prompt injection and red team testing, JSON schema assertion design, evaluation harnesses for Claude, Gemini and GPT, practical adoption of AI-assisted testing and code review inside engineering teams

PROFESSIONAL EXPERIENCE

Girmiti Softwares Pvt Ltd, Bangalore, Karnataka — **March 2022 to Present**

Software Engineer-Quality Assurance | Payments and FinTech

Gilbarco / Costco (NA): AFD Payment Systems QA — August 2025 to Present

- Ran smoke, regression and sanity testing for production releases deployed across more than 500 Costco fueling stations in North America. No critical defects escaped to production.
- Cut average defect resolution time by 35% by working through host, acquirer and terminal logs layer by layer, pulling them with WinSCP and PuTTY from the Linux GSOM and Omnia board environments.
- Validated more than 15 payment and loyalty workflows, including fueling, carwash kiosk, membership billing, deferred payments, QR payments, ShopCards and coupon redemption.
- Tested all three EMV entry modes (contact chip, contactless or NFC, and magnetic stripe) for sale and preauthorisation, including the deferred transaction flows specific to AFD.
- Monitored Jenkins pipelines to trigger regression runs after each deployment and to triage build failures with development. Also ran Zephyr execution cycles with traceability reporting before every production release.

Sensei Sigma (USA): Mobile Trading and In-App Payments QA — April 2025 to February 2026

- Led a QA team of four across the full SDLC for a B2C trading application shipped to both Google Play and the Apple App Store, covering Android, iOS, a React admin portal and the backend API suite.
- Designed and executed REST API test suites in Python and Selenium, automating registration, coin allocation, live market data feeds and the payment processing endpoints inside the app.
- Owned payment testing inside the app end-to-end: card tokenisation, gateway integration, handling of both success and failure, and balance reconciliation after payment, verified in the database with DBeaver.
- Served as primary QA point of contact for the US client, running live feature demonstrations and proof of concepts, taking in change requests, and supporting production after release until the application reached stable status.

Geidea (Saudi Arabia): Settlement and Reconciliation QA March 2024 to April 2025

- Owned QA end-to-end for the settlement and reconciliation platform, following more than 1,000 simulated daily transactions through the full lifecycle from authorisation to final merchant credit.
- Validated the complete merchant onboarding pipeline: merchant creation, POS terminal assignment, MID and TID provisioning, and the first transaction from a newly onboarded terminal.
- Built REST API suites in Postman covering purchase, refund, preauthorisation, reversal and batch settlement across all card entry modes and schemes, reaching over 98% scenario coverage.
- Worked inside a fully integrated acquiring ecosystem spanning more than 20 partner banks. Also triaged live settlement discrepancies in production to limit financial impact on merchants and hold SLA compliance.

Lavego POS: Protocol and Card Network Certification QA March 2023 to February 2024

- Executed L3 card certifications for five international schemes: Visa, Mastercard, Amex, UnionPay and Diners Club. All five passed on the first or second submission cycle.
- Covered the core transaction set on the Lavego POS application, spanning purchase, refund, preauthorisation, reversal, settlement and batch upload, including how the terminal behaves once it loses connectivity to the host.
- Ran precertification of the Lavego POS application against the NEXO protocol and ISO 20022 messaging standards, then secured NEXO certification for Visa and Mastercard, which widened the range of terminals the organisation could certify.
- Operated the UL Certification Tool and ICC Tool to generate certification traces, manage test execution matrices and compile complete evidence packages for scheme submission.

Verifone POS and TMS: Full-Spectrum Deployment Testing August 2022 to February 2023

- Performed functional and integration testing of POS applications across purchase, refund, preauthorisation, settlement, reversal and batch upload, including scenarios where the terminal runs offline.
- Tested the Terminal Management System (VHQ) end-to-end across more than 200 deployed terminals, covering registration, remote configuration push and firmware update workflows.
- Executed backend API testing through Postman directly against the payment processor and a host simulator, parsing ISO-8583 traffic at MTI, DE field and bitmap level to confirm processing integrity and the accuracy of response codes.

Odisha Power Transmission Corporation Ltd (OPTCL), Bhubaneswar, Odisha **March 2021 to February 2022**

Graduate Apprentice, Electrical Engineering

One-year graduate apprenticeship in electrical engineering, completed before moving into software quality assurance.

PROJECTS

AegisQA, Open source AI failure triage and self-healing layer for Playwright and pytest 2026

Python, Playwright, pytest, LLM agents, GitHub Actions | github.com/shakti-mohapatra/AegisQA

- Built a pytest plugin that captures the full context when an end-to-end test fails: stack trace, reduced DOM, screenshot and the failing selector. That context goes to an LLM agent, which returns a root-cause analysis, a suggested selector patch and a confidence score.
- Designed the trust boundary deliberately. Patches that clear the confidence threshold come out as a unified diff for review and are never applied automatically, so nothing an AI wrote reaches the selector map without explicit human approval.
- Shipped a suite of 32 tests that runs with no LLM calls and no API key, plus a GitHub Actions workflow that exercises only the capture path. CI stays free of secrets and API cost.

Fintech-AI-Guard, Open source LLM evaluation framework for regulated financial workflows 2026

Python, promptfoo, FastAPI, JSON Schema | github.com/shakti-mohapatra/fintech-ai-guard

- Built a deterministic, repeatable evaluation engine that checks whether an LLM stays inside a fintech compliance and business logic contract. It covers 10 risk categories, among them numeric precision, PII and PCI handling, and consistency of business logic.
- Reached an 88.9% composite pass rate across the baseline evaluation suite, with 100% resistance to prompt injection, 100% JSON schema validation and no PII or PCI leakage detected.
- Applied payments QA discipline to LLM behaviour through YAML scenario definitions, Python assertions, schema contracts and repeatable regression reporting across Claude, Gemini and GPT, with a FastAPI mock API as the system under test.

EDUCATION

GenAI and Agentic AI Master's Program, LearnBay, *In progress* **July 2026 to April 2027**

Includes a Microsoft Azure AI fundamentals certification, to be completed within the program.

Bachelor of Technology, Electrical Engineering, Parala Maharaja Engineering College, BPUT University, Odisha **2020**